

```
USE intership_database;
```

```
CREATE TABLE Department (  
dept_id INT PRIMARY KEY,dept_name VARCHAR(50));
```

```
CREATE TABLE Employee (  
emp_id INT PRIMARY KEY AUTO_INCREMENT,emp_name VARCHAR(50),salary  
DECIMAL(10,2),hire_date DATE,dept_id INT,FOREIGN KEY (dept_id) REFERENCES  
Department(dept_id));
```

-- Department Data

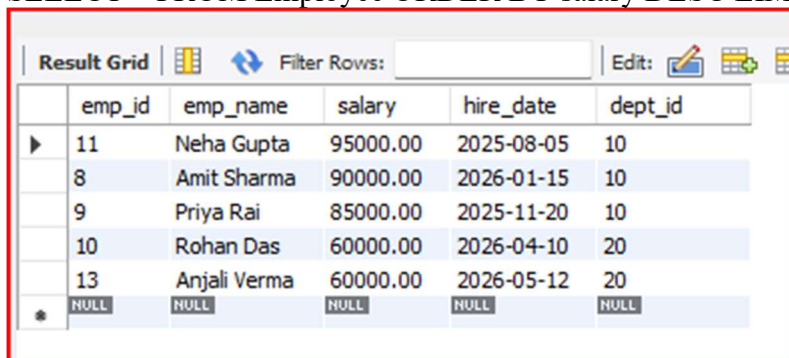
```
INSERT INTO Department VALUES (10, 'IT'), (20, 'HR'), (30, 'Sales');
```

-- Employee Data

```
INSERT INTO Employee (emp_name, salary, hire_date, dept_id) VALUES  
( 'Amit Sharma', 90000.00, '2026-01-15', 10),  
( 'Priya Rai', 85000.00, '2025-11-20', 10),  
( 'Rohan Das', 60000.00, '2026-04-10', 20),  
( 'Neha Gupta', 95000.00, '2025-08-05', 10),  
( 'Vikram Singh', 45000.00, '2026-03-01', 30),  
( 'Anjali Verma', 60000.00, '2026-05-12', 20),  
( 'Amit Sharma', 90000.00, '2026-01-15', 10);
```

1. Top 5 highest salary employees

```
SELECT * FROM Employee ORDER BY salary DESC LIMIT 5;
```

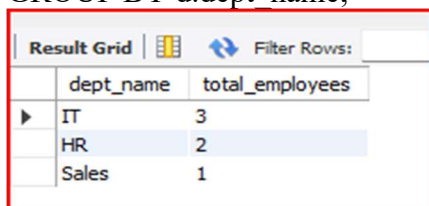


The screenshot shows a database query result grid with the following data:

	emp_id	emp_name	salary	hire_date	dept_id
▶	11	Neha Gupta	95000.00	2025-08-05	10
	8	Amit Sharma	90000.00	2026-01-15	10
	9	Priya Rai	85000.00	2025-11-20	10
	10	Rohan Das	60000.00	2026-04-10	20
	13	Anjali Verma	60000.00	2026-05-12	20
✱	NULL	NULL	NULL	NULL	NULL

2. Department wise employee count

```
SELECT d.dept_name, COUNT(e.emp_id) AS total_employees  
FROM Employee e  
JOIN Department d ON e.dept_id = d.dept_id  
GROUP BY d.dept_name;
```

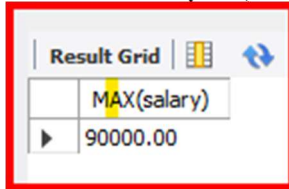


The screenshot shows a database query result grid with the following data:

	dept_name	total_employees
▶	IT	3
	HR	2
	Sales	1

3. Find Second highest salary

```
SELECT MAX(salary) FROM Employee  
WHERE salary < (SELECT MAX(salary) FROM Employee);
```

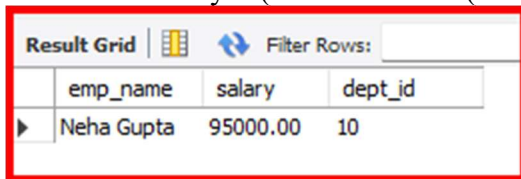


The screenshot shows a 'Result Grid' with two columns. The first column contains the expression 'MAX(salary)' and the second column contains the value '90000.00'.

	MAX(salary)
▶	90000.00

4. Employees whose salary > department average salary

```
SELECT e.emp_name, e.salary, e.dept_id  
FROM Employee e  
WHERE e.salary > (SELECT AVG(salary) FROM Employee WHERE dept_id = e.dept_id);
```

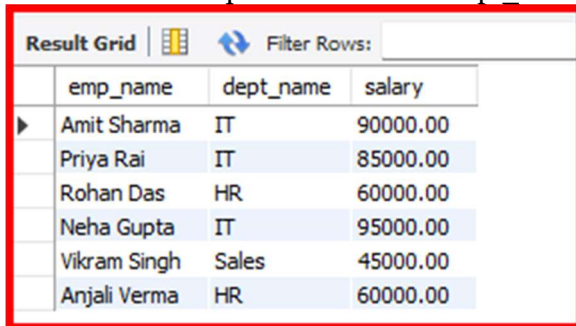


The screenshot shows a 'Result Grid' with four columns: 'emp_name', 'salary', and 'dept_id'. There is one row of data for Neha Gupta.

	emp_name	salary	dept_id
▶	Neha Gupta	95000.00	10

5. Inner Join

```
SELECT e.emp_name, d.dept_name, e.salary  
FROM Employee e  
INNER JOIN Department d ON e.dept_id = d.dept_id;
```

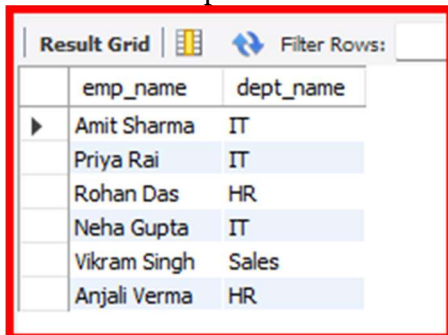


The screenshot shows a 'Result Grid' with four columns: 'emp_name', 'dept_name', and 'salary'. It lists seven employees and their respective departments.

	emp_name	dept_name	salary
▶	Amit Sharma	IT	90000.00
	Priya Rai	IT	85000.00
	Rohan Das	HR	60000.00
	Neha Gupta	IT	95000.00
	Vikram Singh	Sales	45000.00
	Anjali Verma	HR	60000.00

#6. Left Join

```
SELECT e.emp_name, d.dept_name  
FROM Employee e  
LEFT JOIN Department d ON e.dept_id = d.dept_id;
```

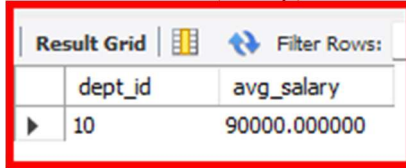


The screenshot shows a 'Result Grid' with two columns: 'emp_name' and 'dept_name'. It lists seven employees and their respective departments, matching the inner join result.

	emp_name	dept_name
▶	Amit Sharma	IT
	Priya Rai	IT
	Rohan Das	HR
	Neha Gupta	IT
	Vikram Singh	Sales
	Anjali Verma	HR

7. Group By with Having

```
SELECT dept_id, AVG(salary) AS avg_salary  
FROM Employee  
GROUP BY dept_id  
HAVING AVG(salary) > 65000;
```

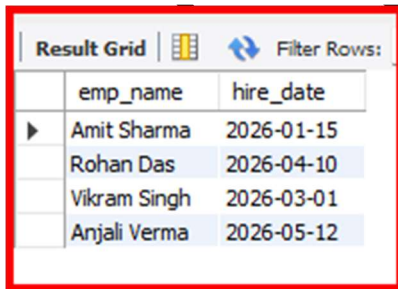


The screenshot shows a 'Result Grid' with two columns: 'dept_id' and 'avg_salary'. There is one row with 'dept_id' 10 and 'avg_salary' 90000.000000. The grid has a 'Filter Rows' button and a 'Result Grid' label.

dept_id	avg_salary
10	90000.000000

8. Employees hired in last 6 months

```
SELECT emp_name, hire_date FROM Employee  
WHERE hire_date >= DATE SUB('2026-06-25', INTERVAL 6 MONTH);
```

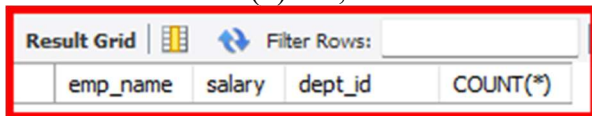


The screenshot shows a 'Result Grid' with two columns: 'emp_name' and 'hire_date'. There are four rows: Amit Sharma (2026-01-15), Rohan Das (2026-04-10), Vikram Singh (2026-03-01), and Anjali Verma (2026-05-12). The grid has a 'Filter Rows' button and a 'Result Grid' label.

emp_name	hire_date
Amit Sharma	2026-01-15
Rohan Das	2026-04-10
Vikram Singh	2026-03-01
Anjali Verma	2026-05-12

9. Find the Duplicates records

```
SELECT emp_name, salary, dept_id, COUNT(*)  
FROM Employee  
GROUP BY emp_name, salary, dept_id  
HAVING COUNT(*) > 1;
```



The screenshot shows a 'Result Grid' with four columns: 'emp_name', 'salary', 'dept_id', and 'COUNT(*)'. The grid has a 'Filter Rows' button and a 'Result Grid' label.

emp_name	salary	dept_id	COUNT(*)
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10. How to remove the duplicates record

```
DELETE e1 FROM Employee e1  
INNER JOIN Employee e2  
WHERE e1.emp_id > e2.emp_id  
AND e1.emp_name = e2.emp_name  
AND e1.salary = e2.salary;
```